Matthew Hockert

PhD Candidate in Applied Economics · University of Minnesota

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Education

PhD in Applied Economics September 2022 – Present
University of Minnesota, Minneapolis, MN

- All but dissertation (ABD), expected graduation December 2026.
- Fifth-year student (2026–2027). Advisor: Elizabeth Davis.
- Research Fields: Program and Policy Evaluation, Public Finance, Housing, Land Use, Transportation.
- Methods: difference-in-differences, regression discontinuity, instrumental variables, forecasting.

Master of Science in Applied Economics September 2020 – June 2024
University of Minnesota, Minneapolis, MN

- Graduate minor in Data Science.
  - Courses: Machine Learning Fundamentals, Principles of Database Systems, Theory of Statistics.
- Thesis: "A Spatial Approach to Agrifood Supply Chain Structure" (Advisor: Hikaru Peterson).
- Recipient of the UMN Applied Economics MS Thesis Award, 2024–25.

B.A. in Economics and Political Science 2017 – 2020
University of Minnesota Duluth

- GPA: 3.7/4.0. *Cum Laude.* University Honors Program, Dean's List.

Professional & Research Experience

Economist (Conservation) – Intern II June 2024 – Present
Hennepin County, Minneapolis, MN

- Evaluate the effects of three Minnesota state programs (Green Acres, Agricultural Preserve, and Rural Preserve) on enrolled and neighboring properties, producing analysis that informs county and municipal program decisions.
- Conservation easement analysis changed Minnesota Statutes 273.117, which now permits metropolitan counties to authorize assessors to account for the impact of conservation easements on property values.
- Built a simulation model in R covering over 427,000 records that projects fiscal outcomes across city, county, and school-district jurisdictions under alternative program enrollment scenarios.
- Built a Quarto document for the cost-of-not-conserving project that layers NLCD land cover, FEMA Ecosystems Services data, and county parcel records to calculate the dollar tradeoffs of conserving versus developing land.
- Automated easement impact reports for municipalities in the county, written for city administrators and county staff.
- Developed and deployed an interactive R Shiny and Leaflet application that lets staff test hypothetical program scenarios on a map and see fiscal outcomes update in real time, used as a decision-support and performance-measurement tool.
  - Designed a scoring algorithm that ranks candidate sites by projected fiscal impact, coordinating the work across county staff and external partners.
  - Layered Metropolitan Council 2040 planned land use and current zoning to quantify foregone future tax capacity across seven cities.
- Automated extraction of terms and conditions from scanned legal documents using natural language processing.

Graduate Research Assistant September 2022 – September 2026
Accessibility Observatory, University of Minnesota, Minneapolis, MN

- Produce the National Accessibility Evaluation, a recurring national study measuring how many jobs are reachable within a given travel time from every census block in the United States.
  - Developed a Python pipeline to pull data from AWS and produce weighted-average job accessibility measures from block to state level across the U.S.
  - Created public datasets on multimodal transportation accessibility (bike, walk, car, transit) using Python and SQL.
- Built R scripts using OpenStreetMap network data and R5 routing engine to calculate ambulance response and transport times to Level I–III trauma centers for every highway segment in rural Minnesota.
- Built data pipelines and ad hoc deliverables for the Metropolitan Council and State Departments of Transportation.
- Developed an internal R Shiny application with Leaflet to display national job accessibility from block group to state level.

Data Science Intern June 2022 – March 2024
Federal Reserve Bank of Minneapolis, Minneapolis, MN

- Built VAR and Kalman Filter forecasting models for macroeconomic stress testing under alternative economic scenarios. The model was delivered into the Bank's production stress-testing system.
- Converted stress-testing production code from MATLAB to R following software development best practices.
- Built R Shiny applications incorporating predictive risk models to identify risk levels by loan and asset type.
- Built R Shiny dashboards using Census ACS and Federal Reserve data to benchmark county-level economic conditions.
- Presented analysis of U.S. banking conditions under CECL policy to examiners and supervisors.

Graduate Research Assistant September 2020 – 2022
Foodshed – Agricultural Commodities Project, University of Minnesota, Minneapolis, MN

- Examined regional food system resiliency during the COVID-19 pandemic using OLS and spatial regression models in R.
- Built comprehensive dataset integrating Census and IMPLAN data.

Undergraduate Research Assistant September 2019 – May 2020
Bureau of Business and Economic Research, University of Minnesota Duluth

- Performed local economic data analysis using IMPLAN, Excel, and SPSS.
- Cleaned and interpreted survey data in Qualtrics, and co-authored publications on the university website.

Technical Skills

Programming:	Python, SQL, R
Data Visualization:	R Shiny, Leaflet, Quarto, Tableau
Statistics & Methods:	Causal inference (difference-in-differences, regression discontinuity, instrumental variables), spatial regression, panel data, forecasting (VAR, Kalman Filter)
Machine Learning:	Supervised learning, decision trees, neural networks, clustering
Geospatial:	ArcGIS, QGIS
Databases & Data Engineering:	Relational database design, DBeaver, AWS
Version Control:	Git, GitHub, GitLab

Research

Dissertation Papers

- "Conservation and Externalities: The Effect of Conservation Easements on Neighboring Residential Parcels"
  - Staggered difference-in-differences on 550,000 residential parcels estimating the trade-off between tax capacity removed and the market value spillovers onto nearby parcels.
- "Rent Control Capitalization into Land Values: Evidence from New Jersey's New Construction Exemption"
  - Triple-difference design on 2024 assessed values exploiting a statutory exemption that excluded newly built 4+ unit buildings from local rent control ordinances. Finds a 4.7% land value premium significant at all three legislative cutoffs.
- "When Do Local Income Taxes Cause Residential Sorting? Evidence from Pennsylvania's Piecewise Earned Income Tax"
  - Continuous stacked difference-in-differences on LEHD/LODES worker flows (2002–2019) identifying a causal effect from Pennsylvania's piecewise local Earned Income Tax.
  - Introduces a binding decomposition isolating the rate changes that move a worker's tax bill, and finds residential sorting away from binding municipal increases.
- "Transportation Changes Land Use: How the St. Croix Bridge Shaped Accessibility, Housing Values, and Travel Behavior"
  - Instrumental-variables difference-in-differences estimating how accessibility gains from a new bridge capitalize into land and improvement values.

Presentations

- "When Do Local Income Taxes Cause Residential Sorting? Evidence from Pennsylvania's Piecewise Earned Income Tax" – Selected Presenter, 20th North American Meeting of the Urban Economics Association (UEA), Chicago, IL, 2026.
- "Transportation Changes Land Use: How the St. Croix Bridge Shaped Accessibility, Housing Values, and Travel Behavior" – Transportation Research Board (TRB) Annual Meeting, Washington, D.C., January 2026.
- "Transportation Changes Land Use: How the St. Croix Bridge Shaped Accessibility, Housing Values, and Travel Behavior" – Transportation Research Conference, Center for Transportation Studies, October 2025.
- "Calculating the Total Travel Time from Crash to Emergency Care in Rural Minnesota" – Transportation Research Conference, Center for Transportation Studies, November 2024.
- "A Spatial Approach to Agrifood Supply Chain Resiliency" – Selected Paper Presenter, AAEA Annual Meeting, July 2023.

Honors and Awards

- UMN Applied Economics MS Thesis Award, 2024–25
- University Honors Program, University of Minnesota Duluth
- *Cum Laude*, Economics Department Honors, Omicron Delta Epsilon, Dean's List (2017–2020)

Volunteer Experience

Coaching Director June 2026 – Present

St. Croix Valley Youth Lacrosse, St. Croix Valley, WI

- Lead coaching for a brand-new youth lacrosse organization, recruiting players and coaches in a community with no established lacrosse tradition.
- Recruit and develop coaching talent, create practice plans used across age levels, and set the standard for a positive team culture across the organization.
- Run clinics to introduce new players to the sport and grow participation in the St. Croix Valley.
- Head Coach, 10U team (2026 season). Head Coach, 8U team (Summer 2025).

Professional Memberships

BUILD Research Network (SSRC) · Agricultural & Applied Economics Association (AAEA)